

Daily Review

Booklet p.10-11 — Getting Around in Space

- 1 Engine off in deep space. Describe the motion.
- 2 Describe the catapult manoeuvre.
- 3 How does an ion drive reach a high speed?
- 4 Why is there no friction in deep space?
- 5 What is thrust?

Constant speed, straight line

Gravity pulls it round, faster

A tiny force, for months

There is no air

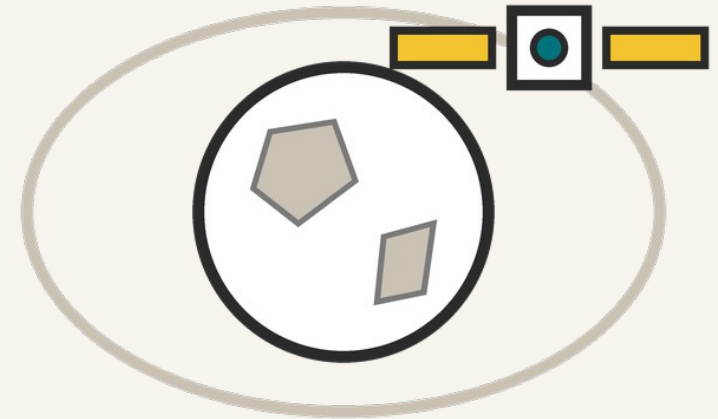
The push from an engine

SPACE

Satellites

Weather, GPS and phone calls — all from objects falling round the Earth.

National 5 Physics (S3)



THE BIG QUESTION

One satellite hangs still above this coast. Another crosses it in ninety minutes. What sets the difference?



Scotland from orbit — NASA

PART 1

What satellites do

What satellites do for us, and why they do not fall down.

By the end of this part...

- 1 I can describe the benefits of satellites
- 2 I can explain how a satellite stays in orbit

Show me — Describe each word

SHOW ME BOARDS

1 satellite

An object in orbit
around another

2 orbit

The path one object
takes around another

3 period

The time for one
complete orbit

4 altitude

The height above the
surface

5 geostationary

Always above the same
point on Earth

Task 1 – what are satellites for?

Booklet p.12

Write down as many uses and benefits
of satellites as you can.

Booklet p.12, Task 1

Task 1 – Answers

ANSWERS

Booklet p.12



GPS



Weather forecasts

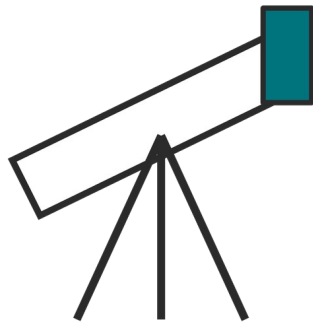


Communications

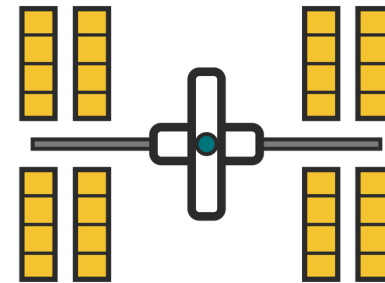
Task 1 – Answers

ANSWERS

Booklet p.12



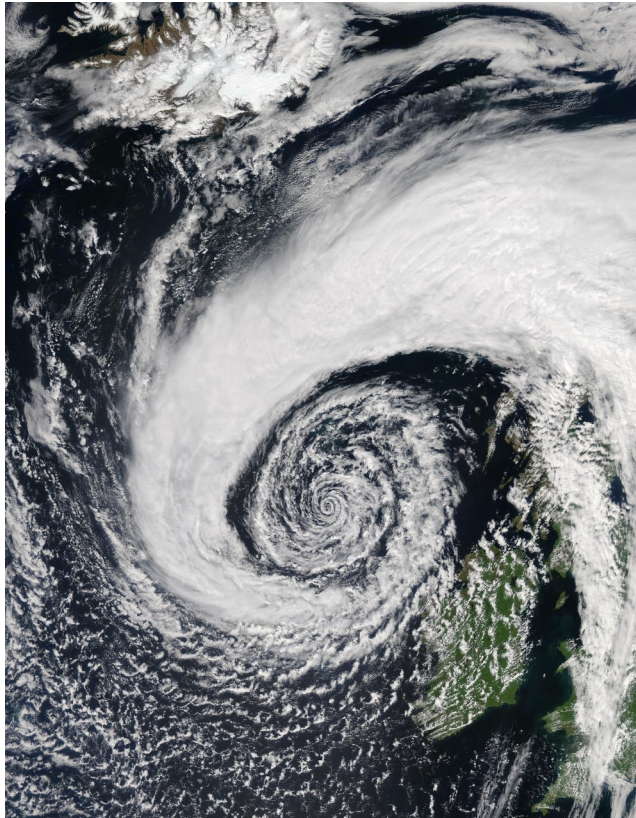
Scientific discovery



Space exploration

Watching the weather

A low pressure system off Ireland



- Satellites photograph the cloud from above.
- This spiral is a deep area of low pressure.
- Forecasters track it as it moves.
- Warnings can come days ahead.

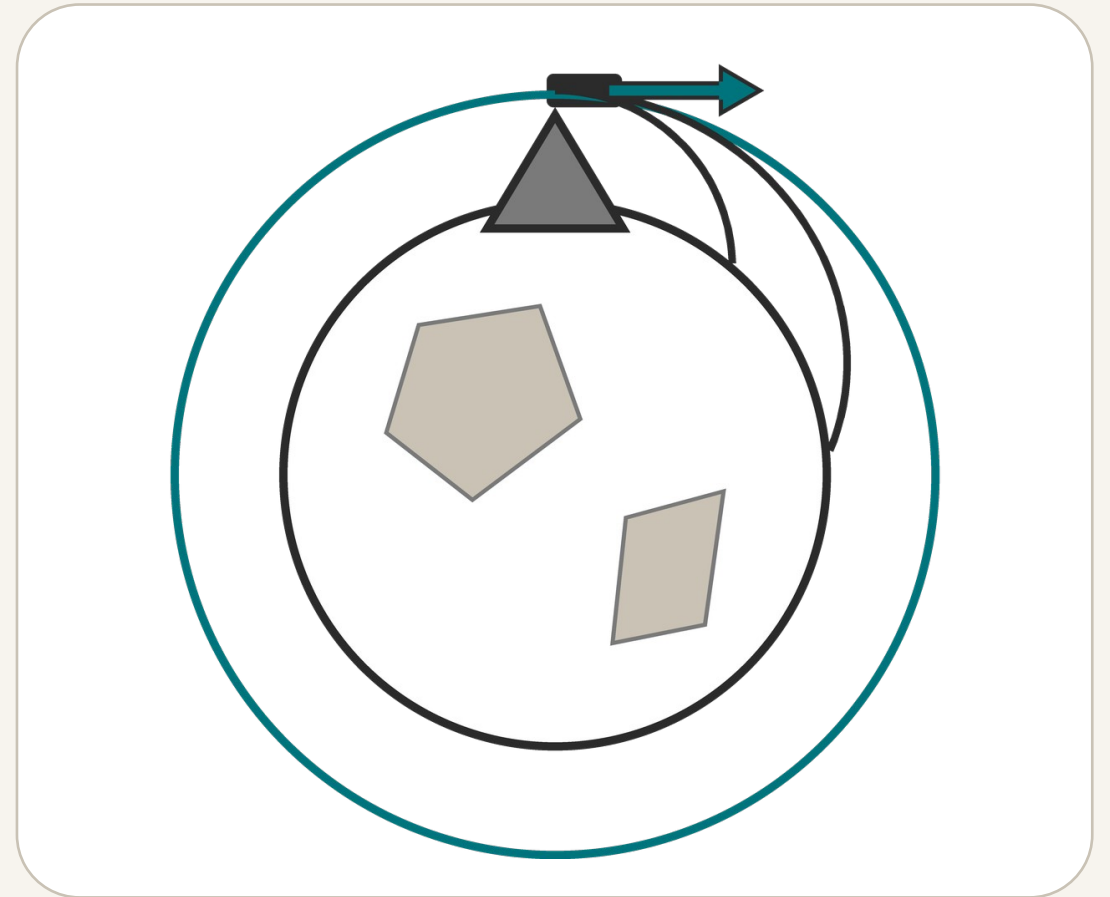


The Whirlpool galaxy — NASA / ESA Hubble Space Telescope

Newton's cannon

Booklet p.12, Task 2

- Fire a cannonball sideways from a very high mountain.
- The faster it goes, the further round it lands.
- Fast enough, the Earth curves away as fast as it falls: an orbit.



Task 2 – Newton's cannon

Booklet p.12

Watch the animation, complete the diagram,
then describe what happens underneath.

Booklet p.12, Task 2

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

A satellite orbits 400 km above the Earth.
Which statement about the satellite is correct?

- A** There is no gravity acting on it
- B** It is falling, and keeps missing the Earth
- C** Its engine keeps it up there
- D** It is beyond the Earth's gravity
- E** It moves in a straight line.

Answer B

Show me — orbits

SHOW ME BOARDS

EXTRA PRACTICE

- 1 What does “period” mean for a satellite?
- 2 What is meant by “altitude”?
- 3 Why does a satellite not crash into the Earth?
- 4 Name a satellite used for scientific discovery.
- 5 How do satellites help forecasters?

The time for one orbit

Its height above the
surface

It falls as the Earth
curves away

Hubble

They photograph the
clouds

Daily Review

Booklet p.12 — Part 1

- 1 Name three uses of satellites.
- 2 Why does a satellite stay in orbit?
- 3 What does the period of a satellite mean?
- 4 What is altitude?
- 5 What is a satellite?

GPS, weather,
communications

It falls as the Earth
curves away

The time for one orbit

Height above the surface

An object orbiting
another

PART 2

Orbits, altitude and period

Choose the height and you have chosen the time for one orbit.

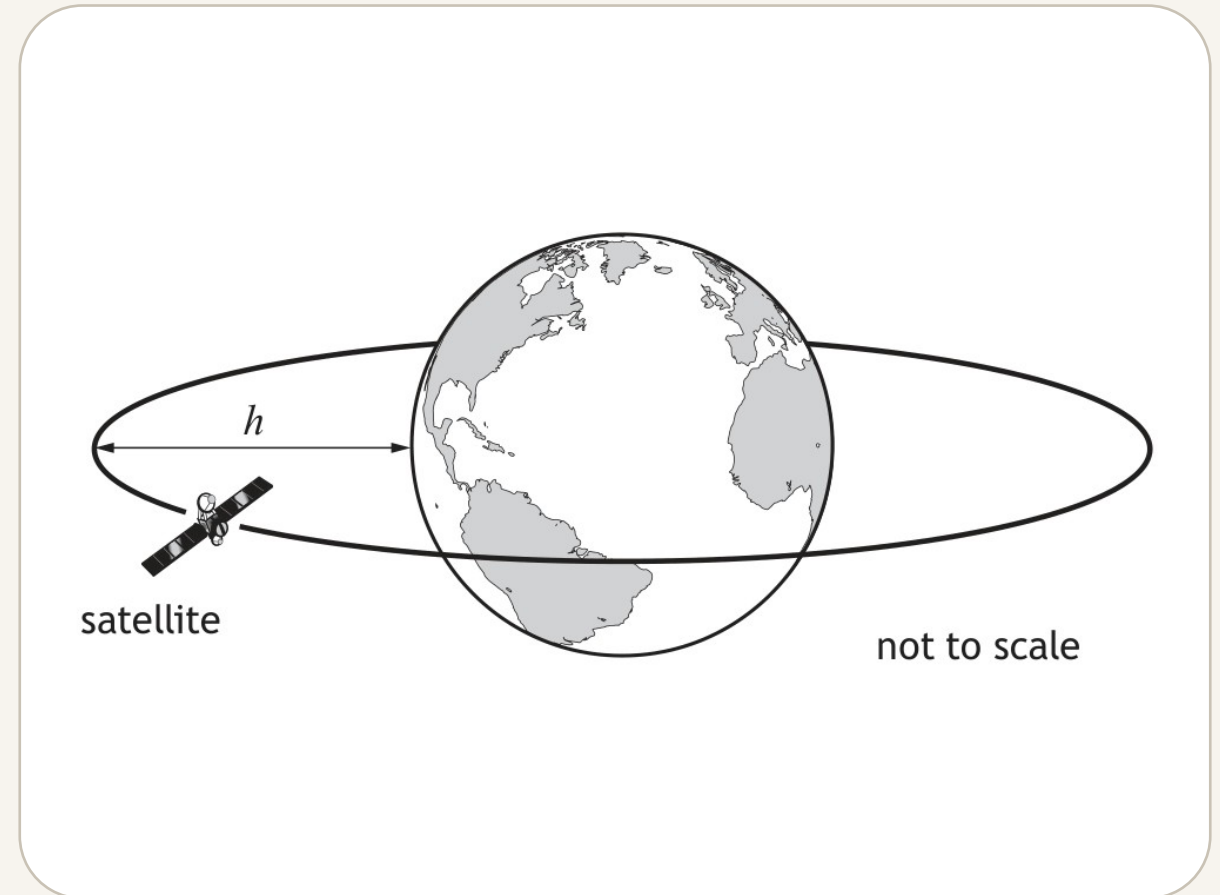
By the end of this part...

- 1 I can state the altitude and period of a geostationary satellite
- 2 I can describe how altitude affects the period of a satellite

Geostationary satellites

Booklet p.12, Task 3

- A period of 24 hours.
- An altitude of 36 000 km.
- It stays above one point on the Earth's surface.



Task 3 – geostationary satellites

Booklet p.12

Complete the definition.

Booklet p.12, Task 3

Task 3 – Answers

ANSWERS

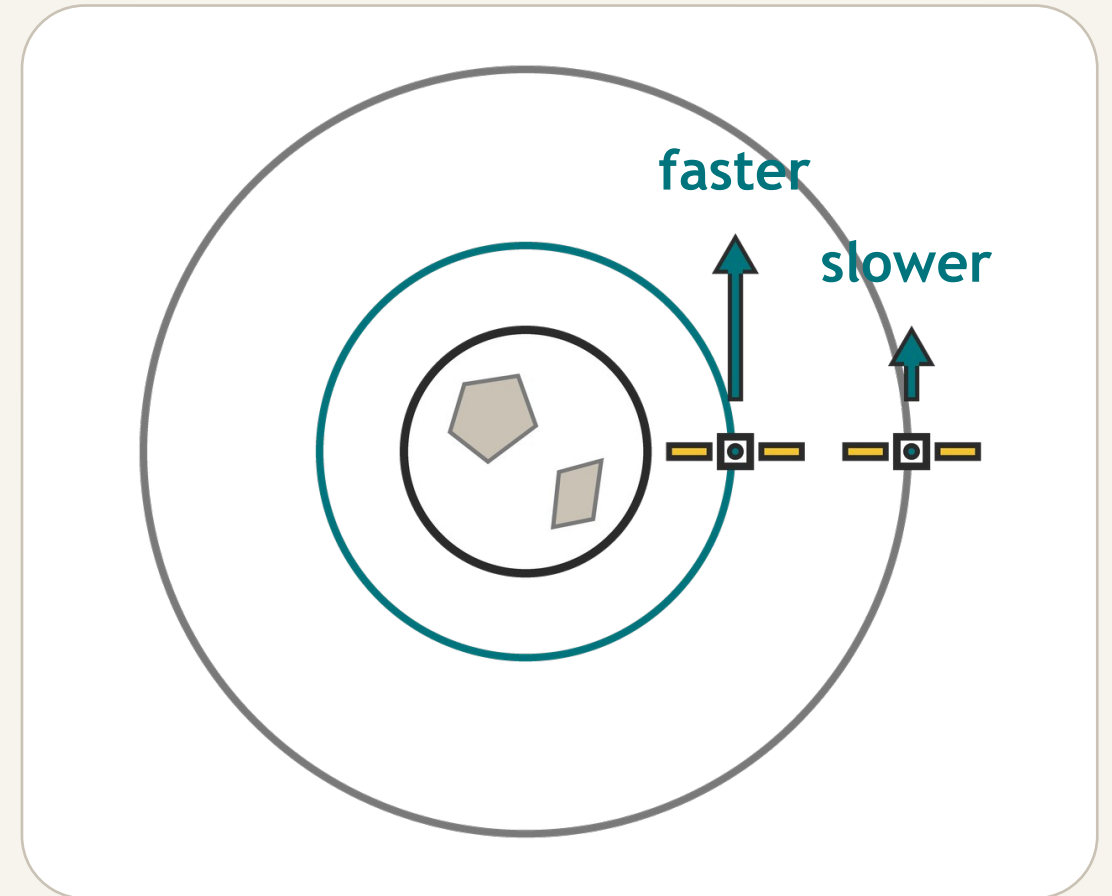
Booklet p.12

Geostationary satellite	
Orbital period	24 hours
Orbital altitude	36 000 km

Why a higher orbit takes longer

Booklet p.12, Task 4

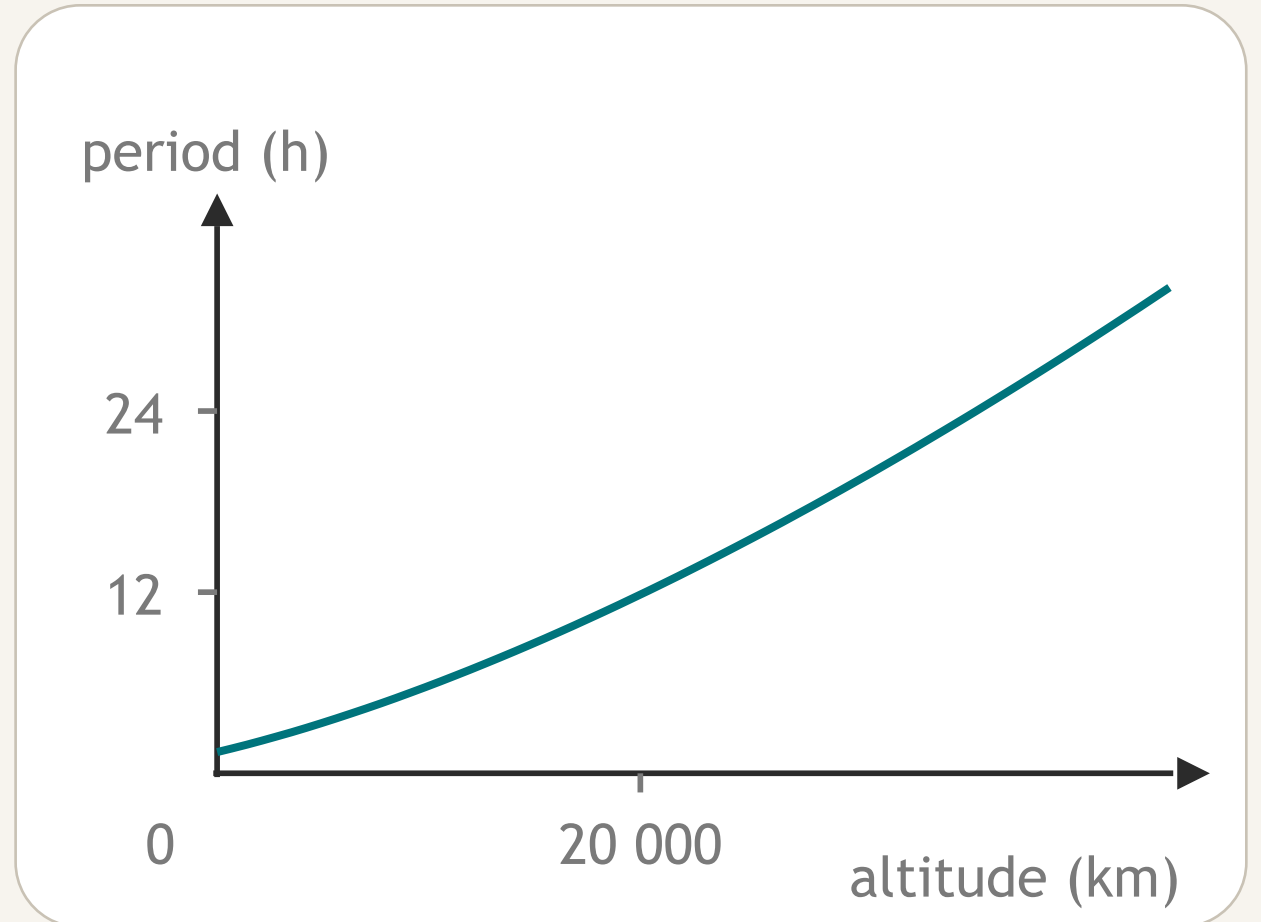
- A higher orbit is a bigger circle: further to travel.
- Gravity is weaker up there, so the satellite moves more slowly.
- Further to go, and going slower — so the period is longer.



Higher orbit, longer period

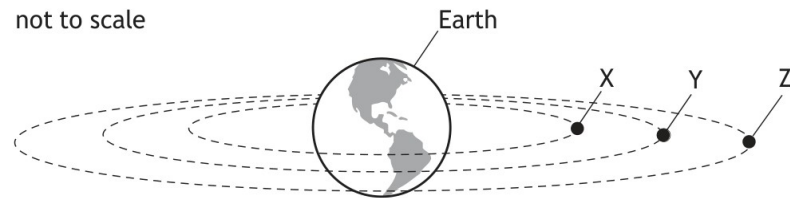
Booklet p.12, Task 4

- The higher the orbit, the longer the period.
- The ISS, about 400 km up: 93 minutes.
- Geostationary, 36 000 km up: 24 hours.



Show me – X, Y and Z

SHOW ME BOARDS



1

Which satellite has the shortest period?

X

2

Which has the longest period?

Z

3

Z is geostationary. What is its period?

24 hours

4

Can X have a period of 24 hours? Why?

No: too low

Task 4 – altitude and period

Booklet p.12

Fill in the blanks.

Booklet p.12, Task 4

Task 4 – Answers

ANSWERS

Booklet p.12

- As the orbital altitude of a satellite increases, the orbital period increases.

Show me — predict

SHOW ME BOARDS

Altitude (km)	Period
700	99 minutes
2000	127 minutes
10 000	5.8 hours

1

Predict the period at 1000 km.

105 minutes

2

Is 20 000 km more than 5.8 hours?

Yes

3

111 minutes: what altitude?

1300 km

4

What pattern does it show?

Higher,
longer

Show me — predict

SHOW ME BOARDS

EXTRA PRACTICE

Altitude (km)	Period
408	93 minutes
1000	105 minutes
36 000	24 hours

1

Estimate the period at 600 km.

97 minutes

2

Is 20 000 km more or less than 24 h?

Less

3

Is the satellite at 408 km geostationary?

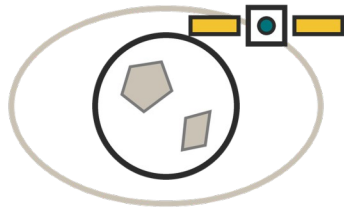
No: 93 min

4

Which satellite goes round most often?

The 408 km one

Scotland in space



UKube-1, Glasgow



SaxaVord, Shetland

- UKube-1, Scotland's first satellite, was built in Glasgow (2014).
- SaxaVord on Unst, Shetland, is licensed as a UK spaceport.

Fly the orbits yourself

TRY THESE

NASA Eyes on the Solar System: fly to any planet.
PhET Gravity and Orbits: change the speed and watch.
Try Newton's cannon: how fast is fast enough?

Extension – BBC Bitesize

Booklet p.13

Revise space exploration on
BBC Bitesize.

Booklet p.13, Extension



BBC Bitesize

Checkpoint

- 1 Give three benefits of satellites.
- 2 State the period and altitude of a geostationary satellite.
- 3 Satellite A orbits higher than B. Which has the longer period?

GPS, weather,
communications

24 hours; 36 000 km

A — higher means longer

Exam practice

Past paper and specimen paper

- National 5 Physics 2023, Section 2, question 3(a). 2 marks.
- Question 3(c). 1 mark. Predicting from a satellite table.
- Specimen paper, Section 2, Q2(d)(ii). 2 marks. Why the ISS stays up.
- Specimen paper, Section 2, Q8(a). 3 marks. A GPS signal's distance.
- Then booklet p.14, Task 5.

LOOKING AHEAD

Forces on Rockets

Thrust, weight and Newton's laws — what it takes to leave the pad.

Plenary

- 1 The big question: why does one satellite hang still over Scotland?
- 2 Name three things satellites are used for.
- 3 What is the period of a geostationary satellite?
- 4 What is its altitude?
- 5 What happens to the period as the altitude increases?

It is geostationary: a 24 h period

GPS, weather, communications

24 hours

36 000 km

It increases

Plenary

EXTRA PRACTICE

- 1 What is meant by the period of a satellite?
- 2 Why does a geostationary satellite stay above one place?
- 3 Which has the shorter period – the ISS or a TV satellite?
- 4 Why does a satellite stay in orbit?
- 5 Describe what happens in Newton's cannon.

The time for one orbit

It goes round once a day

The ISS

It falls as the Earth
curves away

Faster shots land further
round

Extension

- 1 Why must a TV satellite be geostationary?
- 2 Suggest why weather satellites are often in low orbits.
- 3 Why does a satellite in a low orbit need to travel faster?

Extension

EXTRA PRACTICE

- 1 Suggest why GPS needs many satellites, not just one.
- 2 Why can the ISS be seen crossing the sky in a few minutes?
- 3 Suggest a problem with 36 000 km for a phone call.

Course specification

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the benefits of satellites: GPS, weather forecasting, communications, scientific discovery and space exploration (for example Hubble telescope, ISS).
- Knowledge that geostationary satellites have a period of 24 hours and orbit at an altitude of 36 000 km.
- Knowledge that the period of a satellite in a high altitude orbit is greater than the period of a satellite in a lower altitude orbit.